

Problem #1

bankAccount.py

```
class Money:
    def __init__(self, amount: int = 0, denomination: str = "Unknown"):
        self.amount = amount
        self.denomination = denomination
```

testMoney.py

```
from bankAccount import Money

def test_default():
    m1 = Money()
    print(f'Action: Invoking the Money class constructor using Money().')
    print("Output:")
    print(f"Amount: {m1.amount}")
    print(f"Denomination: {m1.denomination}")
    print()

def test_amount():
    m2 = Money(100)
    print(f'Action: Invoking the Money class constructor using Money({m2.amount}).')
    print("Output:")
    print(f"Amount: {m2.amount}")
    print(f"Denomination: {m2.denomination}")
    print()

def test_amount_denomination():
    m3 = Money(100, "USD")
    print(f'Action: Invoking the Money class constructor using Money({m3.amount}, "{m3.denomination}").')
    print("Output:")
    print(f"Amount: {m3.amount}")
    print(f"Denomination: {m3.denomination}")

if __name__ == "__main__":
    test_default()
    test_amount()
    test_amount_denomination()
```

```
Action: Invoking the Money class constructor using Money().
Output:
Amount: 0
Denomination: Unknown

Action: Invoking the Money class constructor using Money(100).
Output:
Amount: 100
Denomination: Unknown

Action: Invoking the Money class constructor using Money(100, "USD").
Output:
Amount: 100
Denomination: USD

Process finished with exit code 0
```

Problem #2

Student.py

```
class Student:
    def __init__(
        self, id_number: int = 0, name: str = "Unknown", course: str = "Unknown" ):
        self.id_number = id_number
        self.name = name
        self.course = course

    def __str__(self):
        return f"{self.id_number} - {self.name} - {self.course}"

    def validate_info(self):
        if self.name.replace(" ", "").isalpha() and len(str(self.id_number)) == 9:
            print("Student information is valid.")
        else:
            print("Student information is not valid.")
```

testStudent.py
from student import Student

```

def main():

    print("Action: Invoking __str__() method with the following Student information:")

    s1 = Student(123456789, "John Doe", "Computer Science")

    print("ID:", s1.id_number)

    print("Name:", s1.name)

    print("Course:", s1.course)

    print("\nOutput:")

    print(s1)


    print("Action: Invoking __str__() method with the following Student information:")

    s2 = Student(12345, "Jane Doe", "Mathematics")

    print("ID:", s2.id_number)

    print("Name:", s2.name)

    print("Course:", s2.course)

    print("\nOutput:")

    print(s2)


    print(
        "Action: Invoking validate_info() method with the following Student information:" )

    s3 = Student(987654321, "Alice123", "Physics")

    print("ID:", s3.id_number)

    print("Name:", s3.name)

    print("Course:", s3.course)

    print("\nOutput:")

    s3.validate_info()
    if __name__ == "__main__":

```

main()

```
Action: Invoking __str__() method with the following Student information:
ID: 123456789
Name: John Doe
Course: Computer Science

Output:
123456789 - John Doe - Computer Science
Action: Invoking __str__() method with the following Student information:
ID: 12345
Name: Jane Doe
Course: Mathematics

Output:
12345 - Jane Doe - Mathematics
Action: Invoking validate_info() method with the following Student information:
ID: 987654321
Name: Alice123
Course: Physics

Output:
Student information is not valid.
```